

A large flock of pink flamingos is gathered in a shallow body of water. The flamingos are densely packed in the foreground and middle ground, with some standing and others wading. The water is a light blue-grey color. In the background, the horizon is visible under a clear blue sky. Two flamingos are seen flying in the upper right portion of the sky.

Collective behavior

IB 431 – Behavioral Ecology
Fall 2025
October 1st 2025

Outline

- I. Quantifying collective animal behavior
- II. Mechanisms driving collective behavior
- III. Broader applications of collective behavior
- IV. Discuss Pratt et al 2002

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Collective animal behavior



Quantifying collective behavior

- **Movement**
 - Cohesion / coordination





Quantifying collective behavior

- Movement
 - Cohesion / coordination
- **Task performance**
 - Nest building / construction, foraging, parental care



Termite mounds

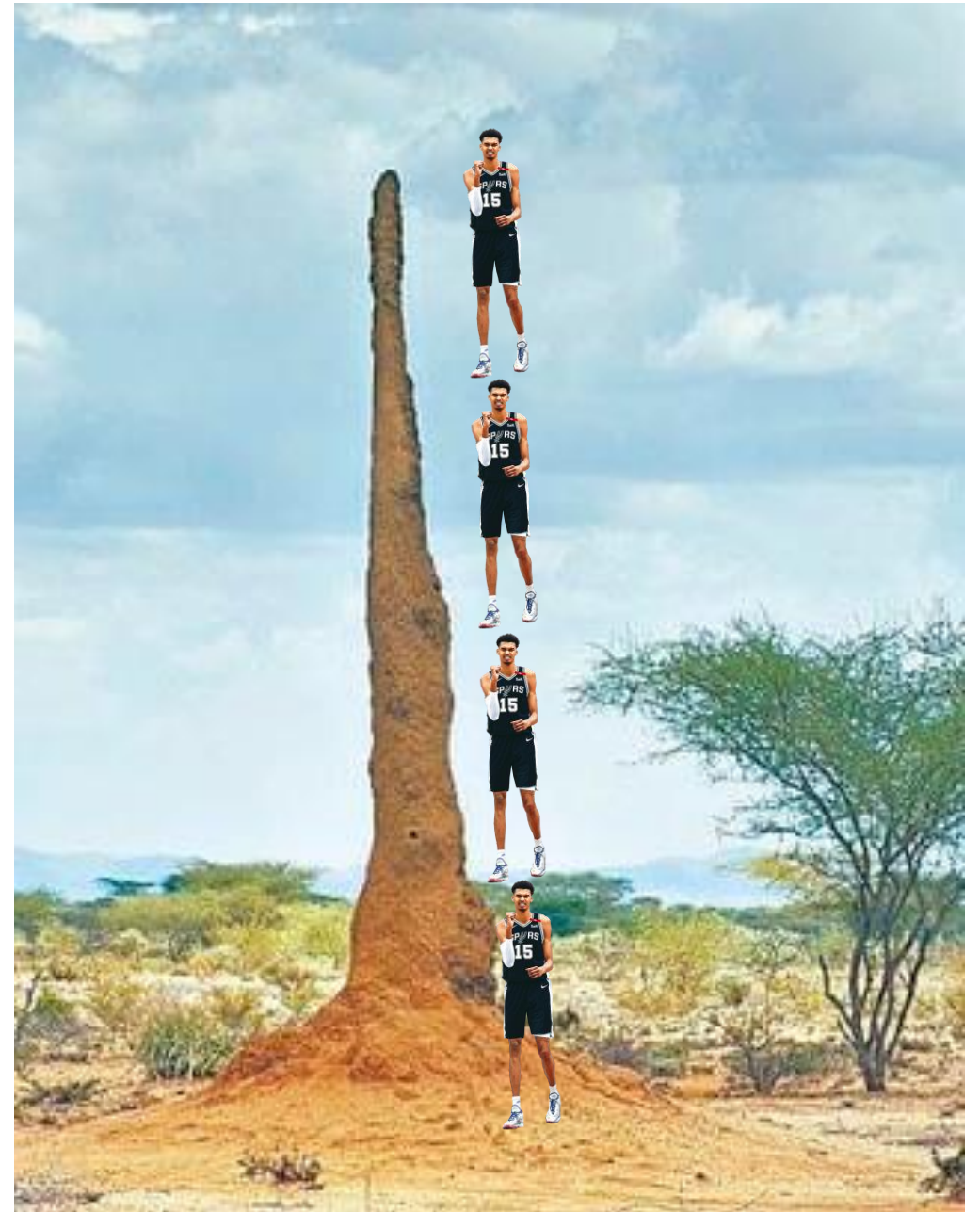
- Largest built structure relative to body size of any organism
 - Up to 9m high



Termite mounds

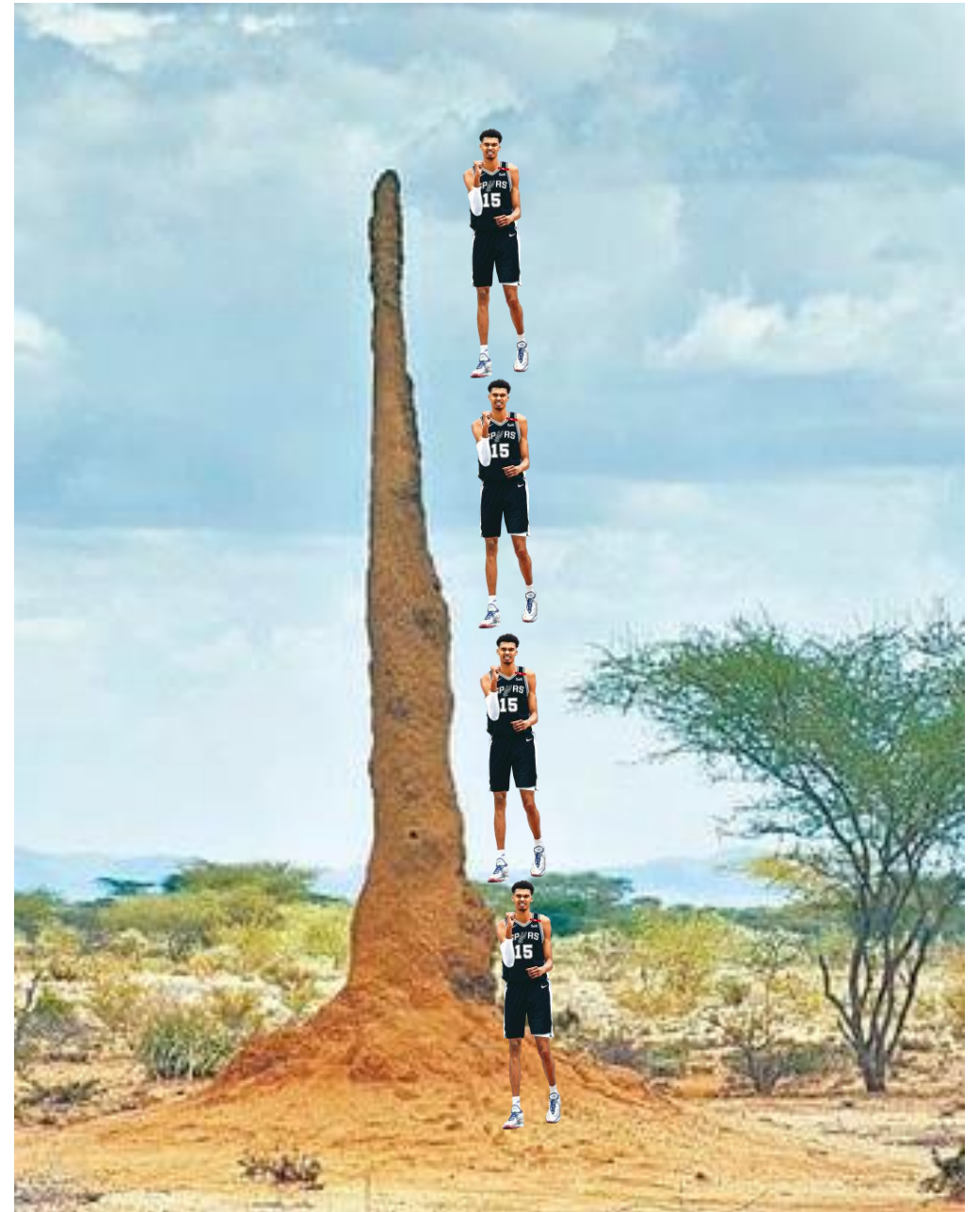


Victor Wembanyama (7'4")



Termite mounds

- Largest built structure relative to body size of any organism
 - Up to 9m high
 - **Equivalent to 3,000 m human building (Burj Khalifa ~830 m)**



Ants can build living bridges or rafts





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0:05 / 1:07



HD



Quantifying collective behavior

- Movement
 - Cohesion / coordination
- Task performance
 - Nest building / construction, foraging, parental care
- **Interactions**
 - Social networks
 - Division of labor / social roles
 - Collective decision making



Social networks in bottlenose dolphins

- Tracked individuals across entire life
- Dolphins have unstable social networks as juveniles, then consistent, stable associations later in life



Galezo et al 2020

Ecological context of collective behavior



- Foraging
- Avoiding predation
- Courtship / parental care
- Communication / transfer of information
- Spread of pathogens

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Collective animal behavior

- Coordinated behavior of animal groups
 - **Individuals follow local interaction rules**



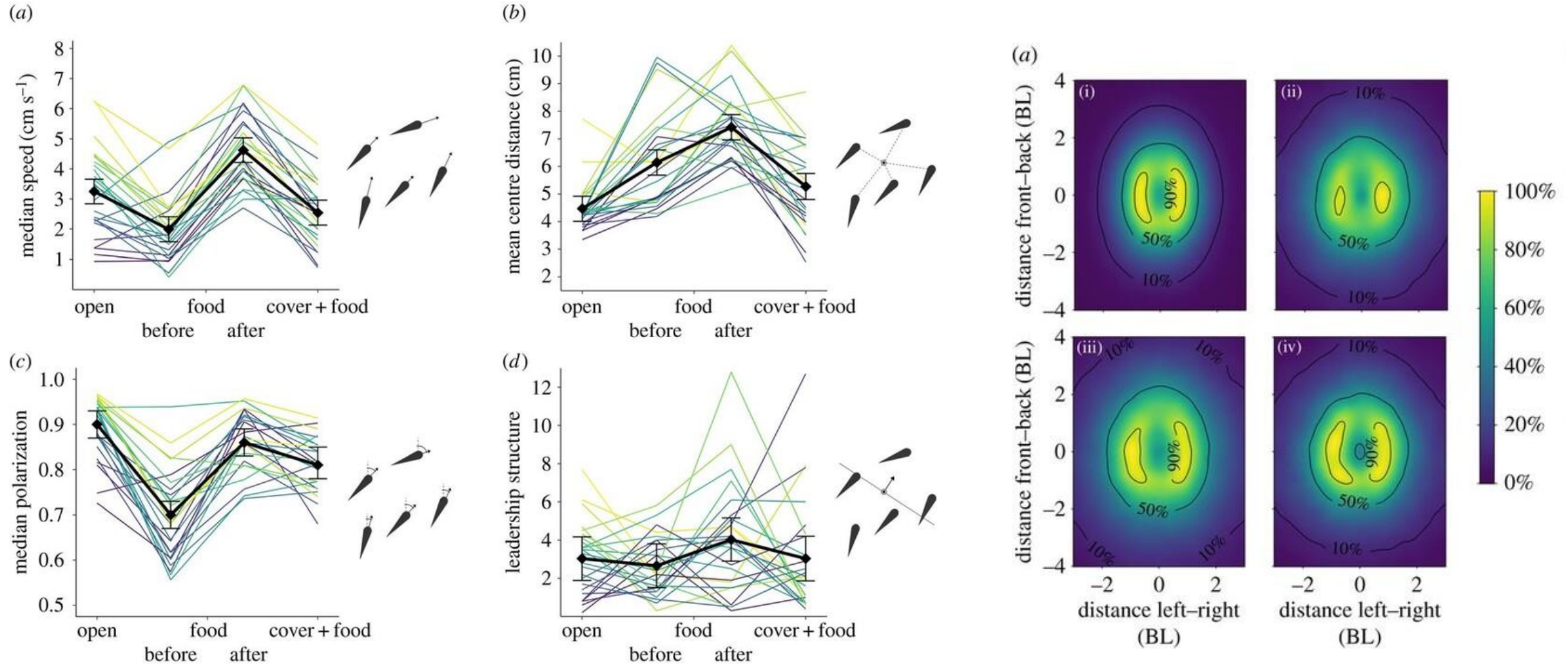


Local decision rules in fish schools

- Zone of attraction: avoid getting too isolated from neighbors
- Zone of repulsion: move away from neighbors that are too close
- Zone of alignment: adopt same orientation as neighbors



Fine-scale movement data



Other examples of local decision rules

- Quorum rules – select behavior once certain threshold of others are performing it
- Random movement until you encounter something
- Feedback loops (+ or –)
- Time threshold for given behavior



Leaders take turns in V formation

Collective animal behavior

- Coordinated behavior of animal groups
 - Individuals follow local interaction rules
 - **Local interactions can generate emergent outcomes**



Emergent outcomes

- Group level behaviors that would not be predicted by individual interactions alone
- Decentralized = no leadership, behavior emerges from local interactions alone
- Fish schools, bird flocks, locust swarms, etc. are examples of emergent outcomes



Collective decision-making and emergent outcomes

- Collective cognition – groups may achieve tasks that individual cannot

Collective decision-making and emergent outcomes

- “Many wrongs” principle
 - Variation among individuals leads to convergence on optimal decision
 - “accuracy of migrating birds is greater than predicted from individual navigational error rates”



Simons 2004

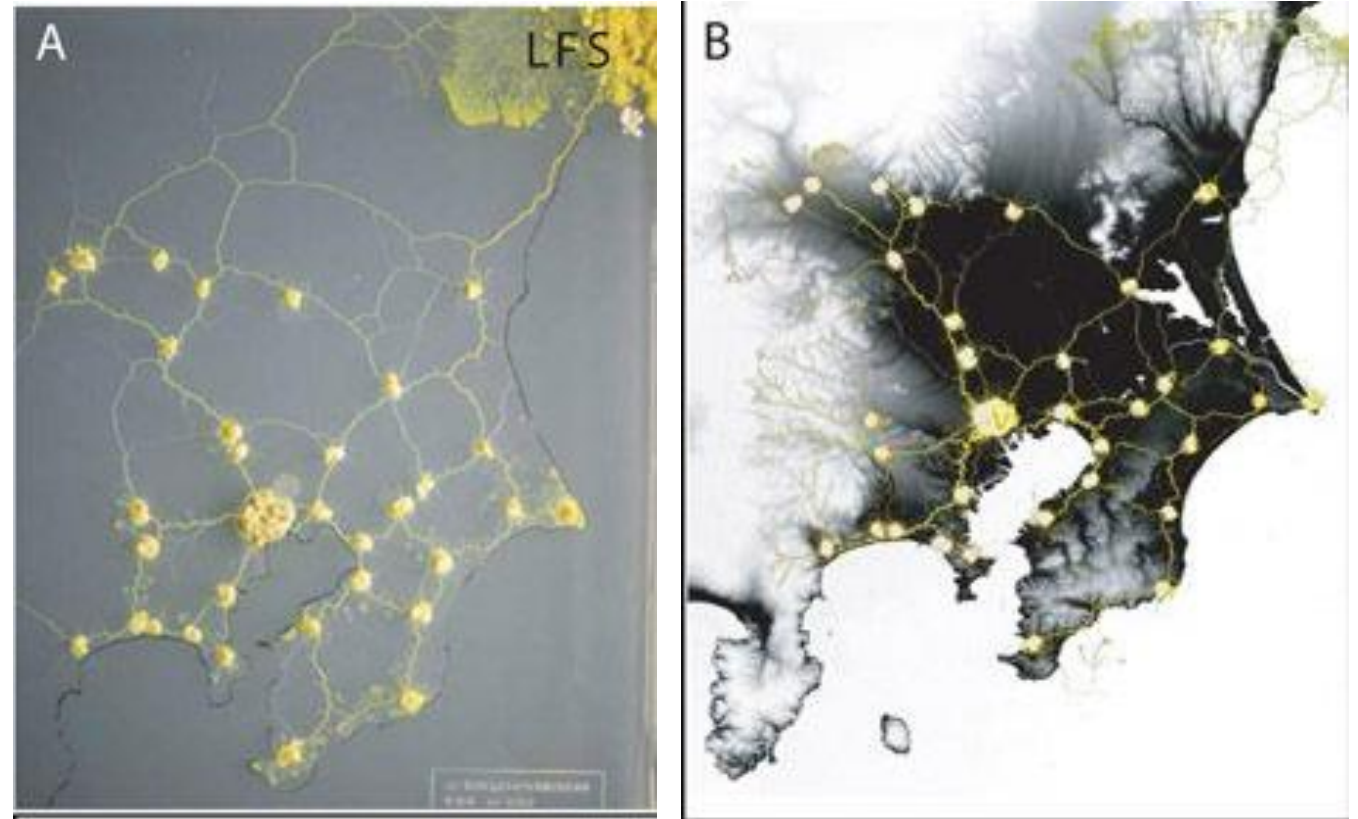
Collective decision-making and emergent outcomes

- Slime molds
 - Single-celled
 - Form colonial “plasmodia”



Collective decision-making and emergent outcomes

- Slime molds
 - Presented with food sources arranged like the city of Tokyo
 - Built network resembling Tokyo rail system



Jabr & Rothschild 2012

Colonial organisms

- **Siphonophores**
 - Portuguese Man-o-War

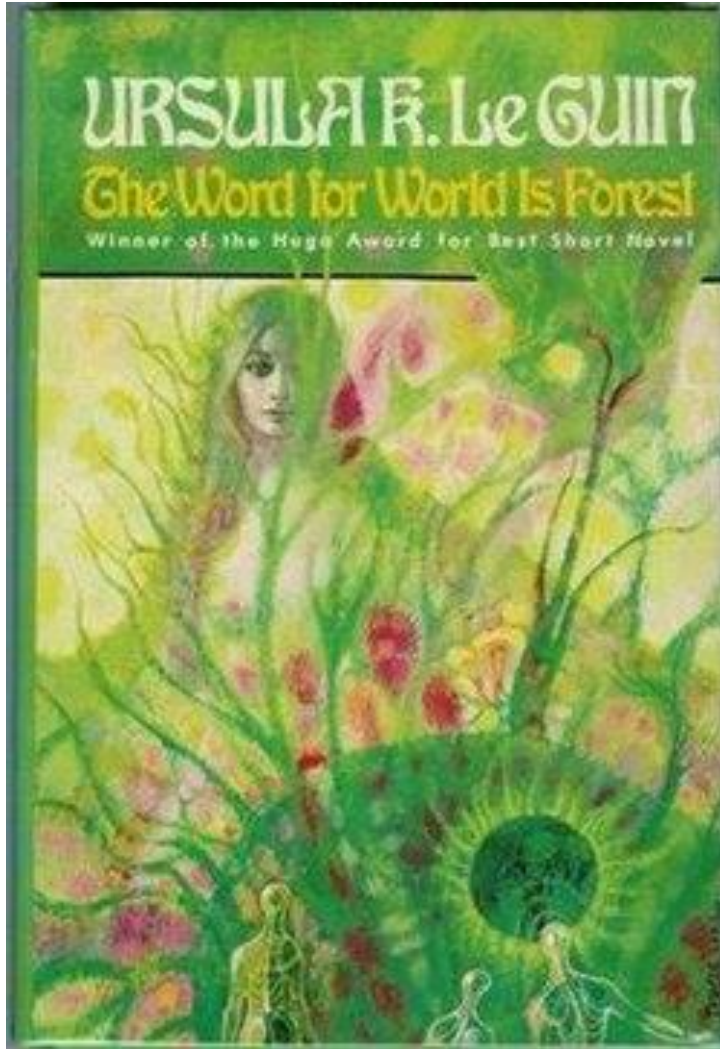


Colonial organisms

- Siphonophores
- **Aspen trees**
 - Utah – organism made up of 47,000 tree trunks all connected by a single root system



Colonial organisms



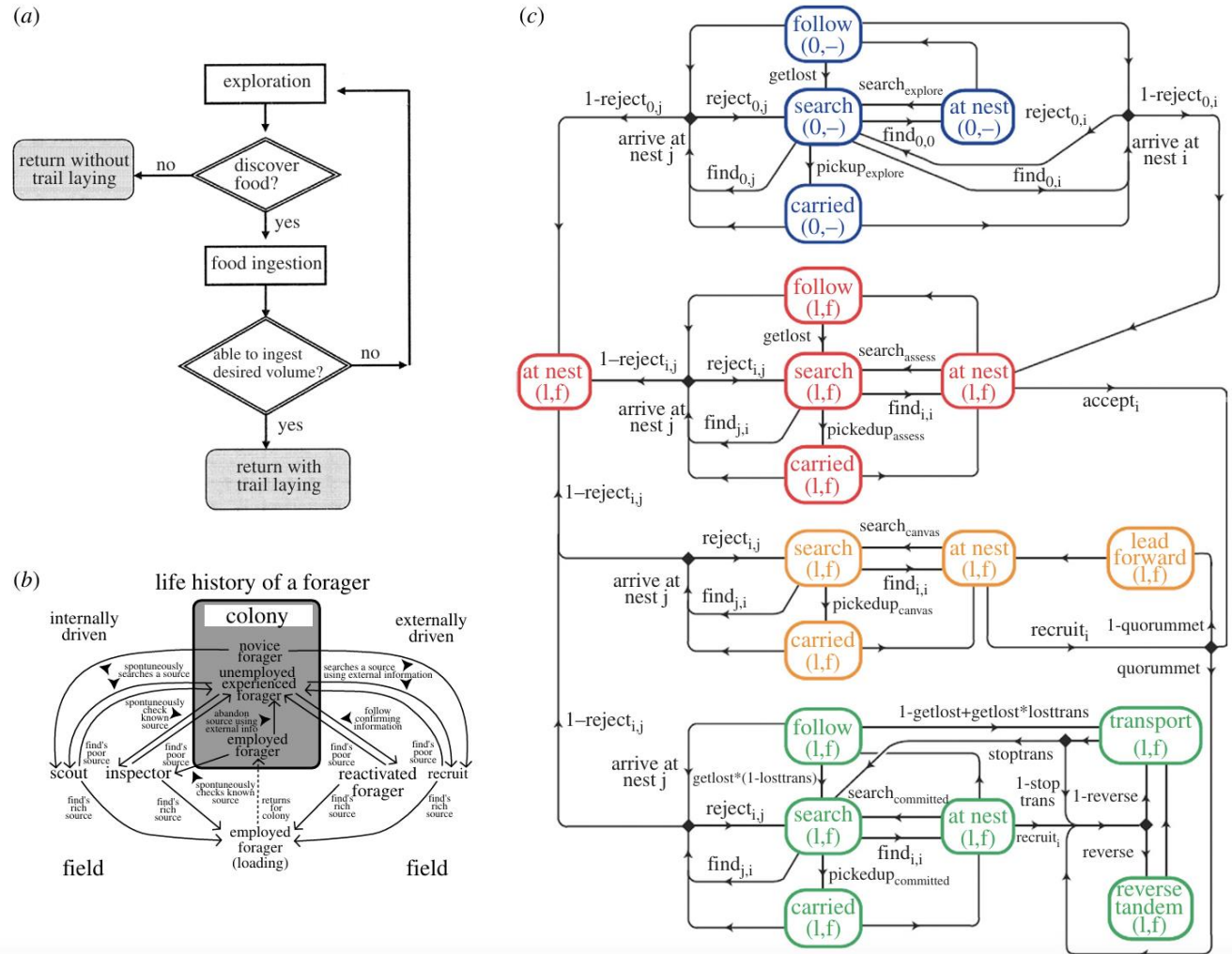
Colonial organisms

- Siphonophores
- Aspen trees
- **Honey mushroom**
 - Oregon – largest organism in the world



Mathematical modeling of collective behavior

- Field of collective behavior built on modeling
 - Information transfer
 - Decision making algorithms
 - Optimality models
 - Simulations



Outline

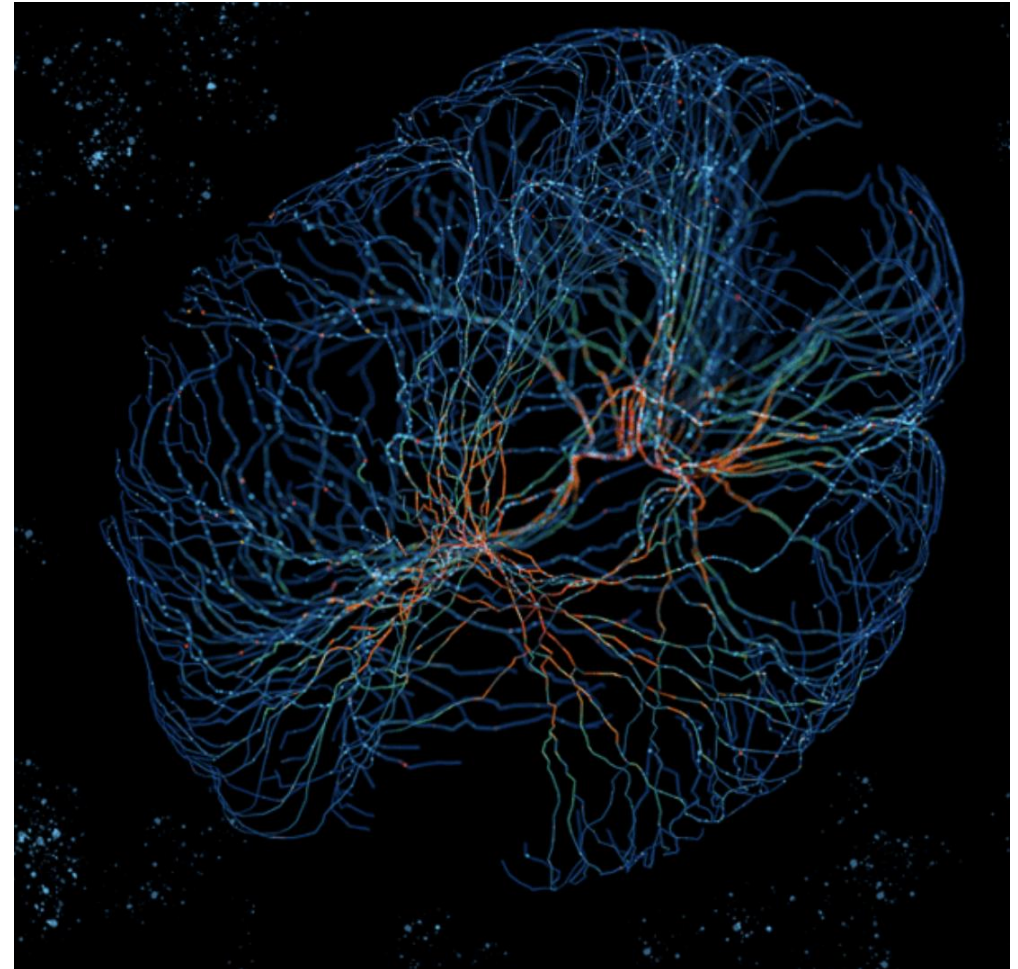
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Applications of collective behavior

- Concepts of emergent outcomes, individual decision rules, etc. apply to many fields

Applications of collective behavior

- **Neural networks**



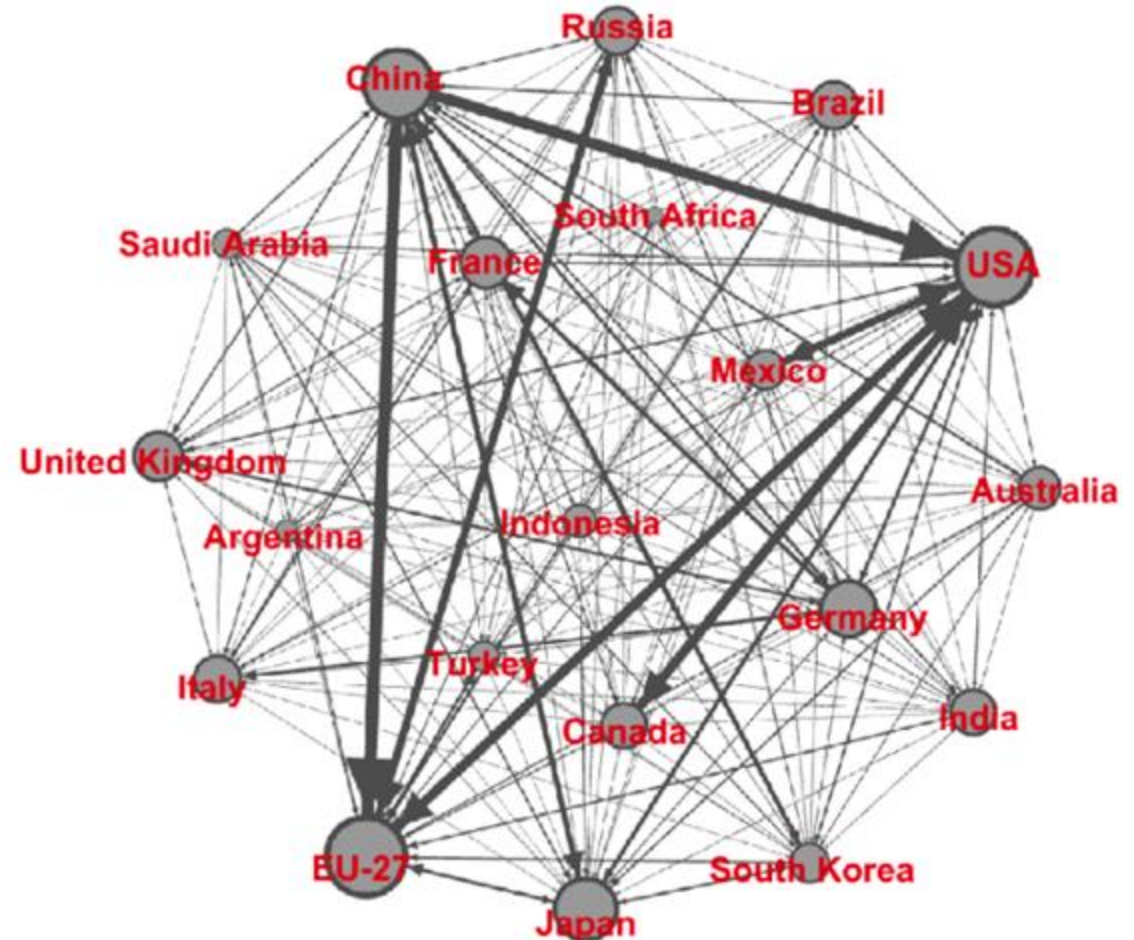
Applications of collective behavior

- Neural networks
- **Human behavior and crowds**



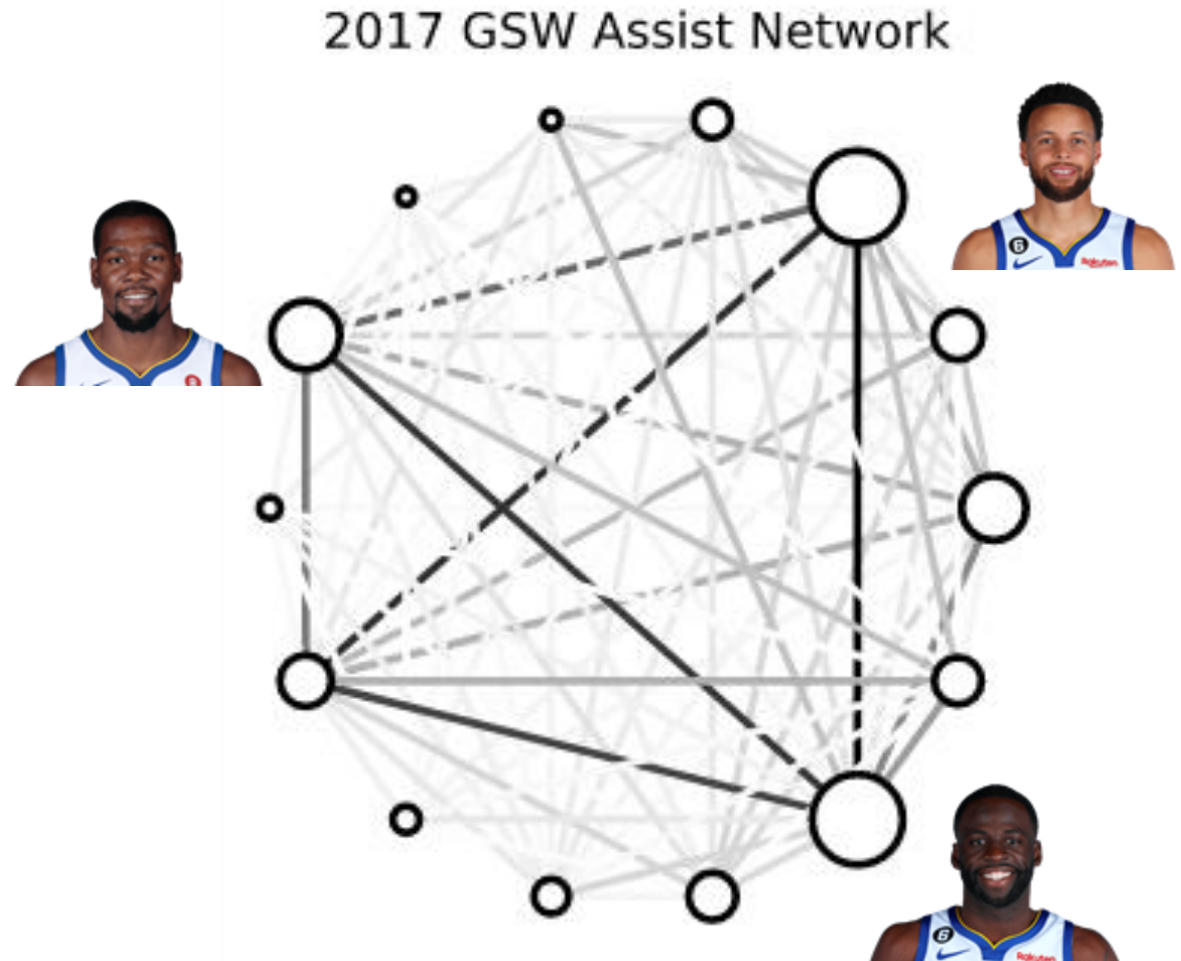
Applications of collective behavior

- Neural networks
- Human behavior and crowds
- **International trade**



Applications of collective behavior

- Neural networks
- Human behavior and crowds
- International trade
- **Assessing offensive efficiency in basketball**



Fakoya et al 2022

Applications of collective behavior

- Neural networks
- Human behavior and crowds
- International trade
- Assessing offensive efficiency in basketball
- **Robotics**



Research gaps in collective behavior

- Ecological divergence of collective behavior in response to climate change?
- Role of individual differences on collective outcomes?
- Genetic mechanisms of collective behavior?



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